



ST. VINCENT PALLOTTI COLLEGE OF ENGINEERING & TECHNOLOGY, NAGPUR

(An Autonomous Institution Affiliated to Rashtrasant Tukadoji Maharaj Nagpur, University)

Department Of Computer Science and Business Systems

Major Project Seminar

On

***Real-Time Anomaly Detection and Semantic Event Mapping in
Financial Time Series***

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Introduction

The Data Explosion

Modern financial markets generate massive amounts of data every millisecond, split into structured data (price, volume, order books) and unstructured data (news, SEC filings, social media).

The Evolution of Trading

While institutional algorithmic trading systems can process and react to both data types instantaneously, retail trading platforms remain heavily focused on structured charting.

The Project Focus

This project bridges the gap between text-based global events and time-series financial charts. It aims to build a system that not only detects when a market anomaly occurs but autonomously explains why it occurred in real-time.



Problem Statement

Information Asymmetry and Cognitive Overload in Active Trading and Investment

The Bottleneck

When sudden, high-volatility price fluctuations occur, traders must manually cross-reference charts with news feeds to understand the catalyst. By the time a human reads the news, filters the noise, and maps it to the chart, the market opportunity has passed or the risk has materialized.



Why it is Important

Milliseconds matter in market reactions. Unexplained volatility leads to uninformed risk-taking.

Real-World Relevance

The rise of meme stocks, algorithmic flash crashes, and rapid earnings reactions have made markets highly sensitive to instantaneous news drops. Retail traders lack the low-latency context needed to navigate these events.

Proposed Title

Real-Time Anomaly Detection and Semantic Event Mapping in Financial Time Series

Scope of the Project

Focused on US Equities (or major Indian NSE stocks depending on API availability).

Ingesting live tick/minute-level market data and real-time news feeds.

Applying NLP for sentiment analysis and machine learning for anomaly detection.

Outputting a synchronized, low-latency visual chart.

Explanation of Title

Refers to the market edge or excess return gained from superior information.

Context

Providing the qualitative reasoning (the "why") behind the quantitative price action.

Objectives of the Project

Specific Goals

Build a low-latency data ingestion engine capable of processing real-time WebSockets.

Implement machine learning models (e.g., Isolation Forests) to flag price/volume anomalies.

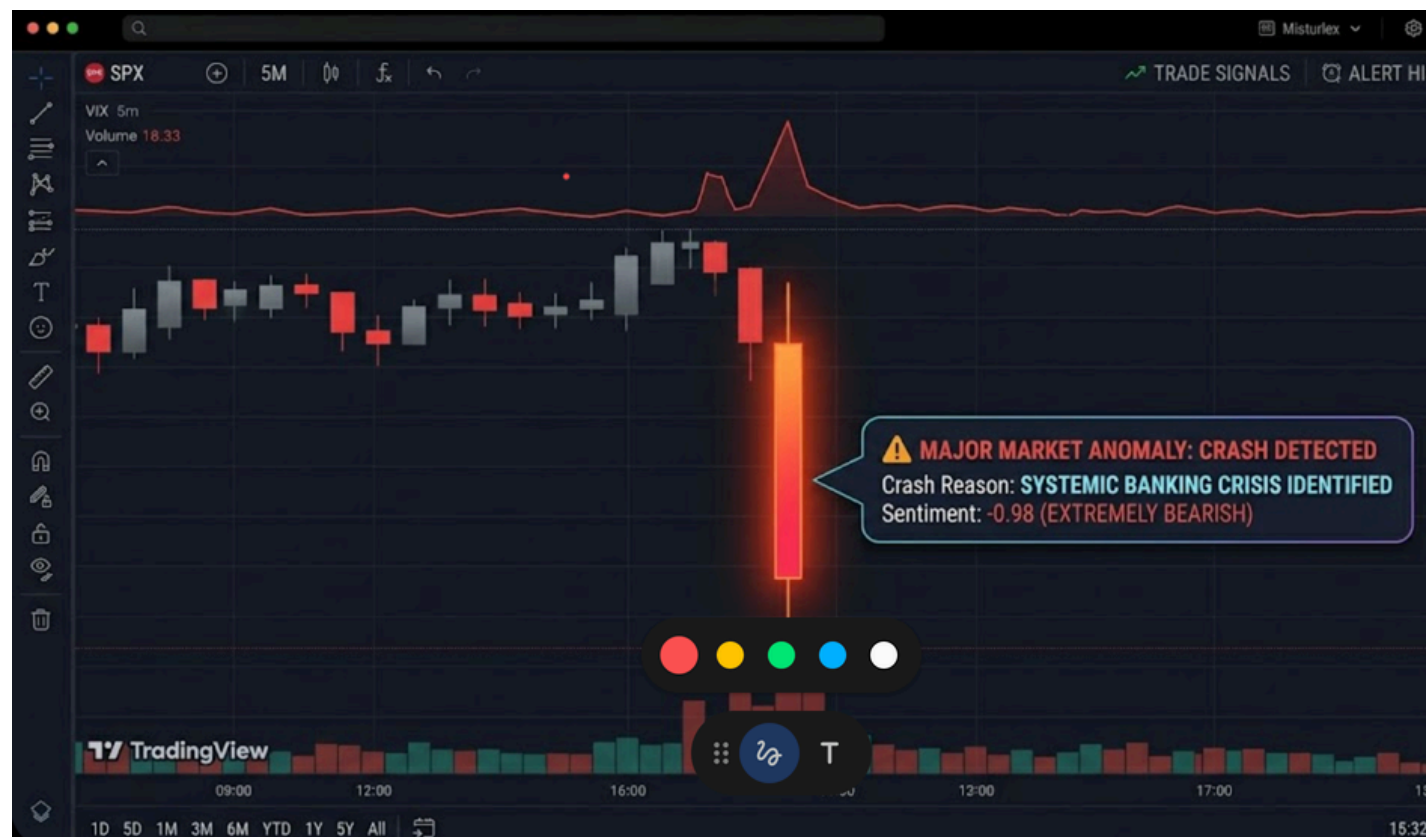
Deploy a fine-tuned financial NLP model (like FinBERT) to extract event entities and sentiment.

Main Objective

To engineer an automated pipeline that detects statistical anomalies in financial time-series data and instantaneously maps them to contextual, sentiment-analyzed news events.

Expected Outcomes

A functional, dark-mode web dashboard featuring interactive candlestick charts where statistically significant candles are automatically annotated with actionable, NLP-derived news insights.



Literature Survey

Existing Systems: Current research successfully applies Deep Learning (LSTM) for price prediction and NLP (BERT) for sentiment analysis independently.

Limitations of Current Solutions: Most systems operate on delayed, daily timeframes (EOD data) rather than intraday tick data. They process text in silos and lack a unified, low-latency visual mapping system.

Gap Identified: There is a critical lack of real-time tools that bridge the gap between quantitative anomaly detection and qualitative NLP event attribution within a single, millisecond-responsive retail trading interface.

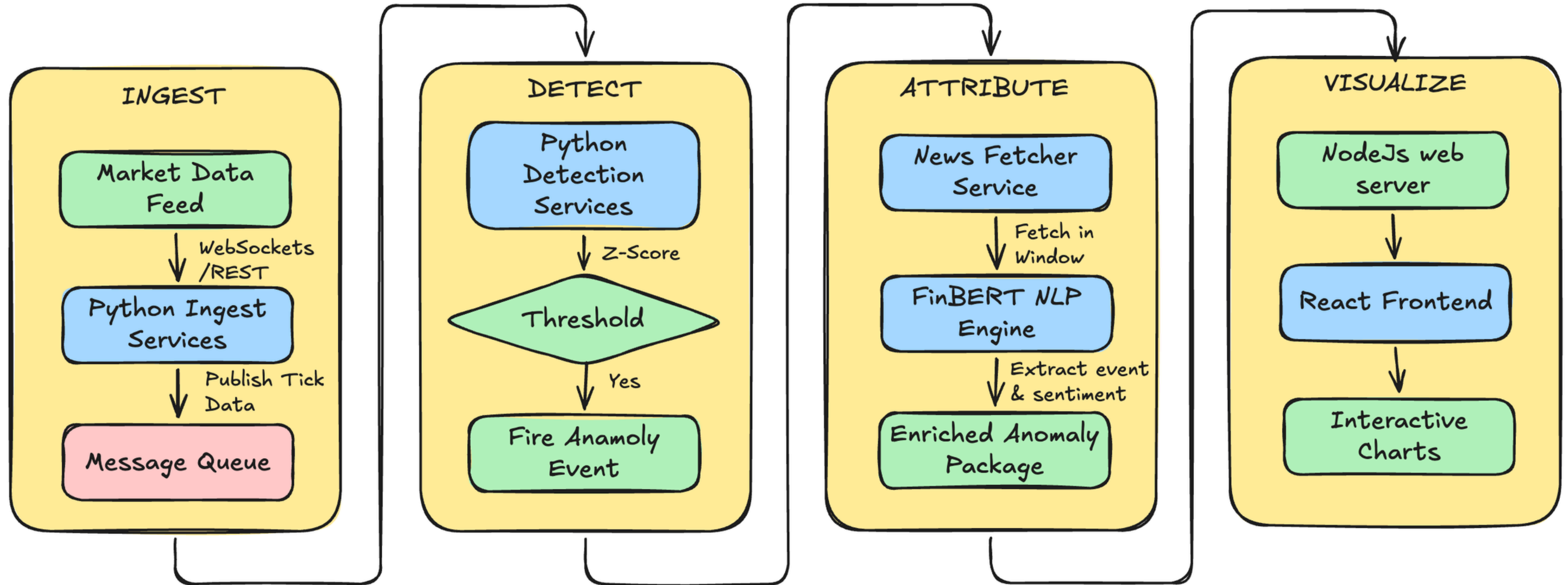
Sr. No.	Paper Title and its Author	Details of Publication with Year	Methods Used	Findings	Limitation
1	FinBERT: Financial Sentiment Analysis with Pre-trained Language Models (D. Araci)	arXiv, 2019	BERT, Transfer Learning	Outperforms traditional ML in financial sentiment tasks.	Focuses solely on text; not linked to live trading charts or price data.
2	Deep Learning for Event-Driven Stock Prediction (X. Ding et al.)	IJCAI, 2015	Neural Tensor Networks, CNN	Extracting structured events from news improves price prediction.	High latency; operates on daily timeframes rather than intraday anomalies.
3	Financial Sentiment Analysis Using FinBERT with Application in Predicting Stock Movement (X. Zhang et al.)	arXiv, 2023	FinBERT, LSTM, ARIMA	Sentiment combined with LSTM significantly improves trend forecasting.	Aggregates sentiment daily; misses millisecond-level flash crashes.
4	Real-Time Stock Market Anomaly Detection Using Machine Learning (Y. Ganiyu et al.)	Towards Data Science, 2024	Isolation Forest, Kafka	Highly effective at flagging high-volume intraday anomalies.	Lacks semantic context; detects the anomaly but cannot explain the cause.

Literature Survey

Sr. No.	Paper Title and its Author	Details of Publication with Year	Methods Used	Findings	Limitation
5	An Evaluation of Large Language Models in Financial Sentiment Analysis (A. Dmonte et al.)	IEEE Big Data, 2024	LLMs, FinBERT, OSCAR corpus	Domain-specific models (FinBERT) outperform general LLMs in finance.	Computationally heavy; difficult to deploy for high-frequency, low-latency tasks.
6	Stock Market Anomaly Detection using Isolation Forest and High-Frequency Data (M. Liu et al.)	IEEE Access, 2021	Isolation Forest, Z-Score	Accurately identifies flash crashes in high-frequency order books.	Purely quantitative; ignores external unstructured data entirely.
7	Volatility Prediction using Financial News Sentiment (J. Atkins et al.)	Quantitative Finance, 2018	Naive Bayes, Latent Dirichlet	News sentiment is highly correlated with intraday market volatility.	Uses outdated NLP models (BoW), leading to poor contextual understanding.
8	Algorithmic Trading based on Sentiment Analysis and Anomaly Detection (S. Kumar et al.)	Expert Systems with Applications, 2022	SVM, K-Means	Combining anomalies with sentiment yields higher risk-adjusted returns.	Static dataset testing; lacks a real-time event-mapping UI.
9	A Hybrid LSTM-based Approach for Event-Driven Stock Price Anomaly Detection (R. Chen et al.)	Knowledge-Based Systems, 2023	LSTM, Attention Mechanisms	Attention layers effectively weigh the impact of different news events.	Model is a black box; lacks explainability and visual mapping for the user.
10	Impact of News Sentiment on High-Frequency Stock Price Anomalies (A. Groß-Klußmann)	Journal of Economic Dynamics and Control, 2019	Econometrics, VAR models	Significant price jumps are almost always preceded by specific news.	Retrospective analysis only; cannot be used as an active alerting tool.

Methodology

Approach to Solve the Problem: Event-driven microservices architecture separating heavy data-crunching from high-speed data delivery.



Workflow Diagram

Expected Results

Output of the System

A live financial chart with automated markers highlighting exact moments of abnormal volatility.

Interactive tooltips displaying the extracted news headline, the isolated event (e.g., "Earnings Miss"), and a FinBERT sentiment score (ranging from -1.0 to +1.0).

What you aim to achieve

A seamless dashboard where quantitative market movements and qualitative global news are merged into a single view.

Benefits

Drastically reduces cognitive overload for traders.
Bridges the information asymmetry gap between retail and institutional traders.
Provides a highly valuable dataset for algorithmic trading strategies.

Timeline

Phase 1 (Weeks 1-2):

Literature review completion, API key acquisition (News APIs, Polygon/Binance), and system architecture design.

Phase 2 (Weeks 3-5)

Milestone 1: Develop Rust/Node.js data ingestion pipelines and implement quantitative anomaly detection algorithms in Python.

Phase 3 (Weeks 6-8)

Milestone 2: Integrate FinBERT for NLP sentiment analysis and establish the event-trigger bridge between the anomaly detector and the news fetcher.

Phase 4 (Weeks 9-11)

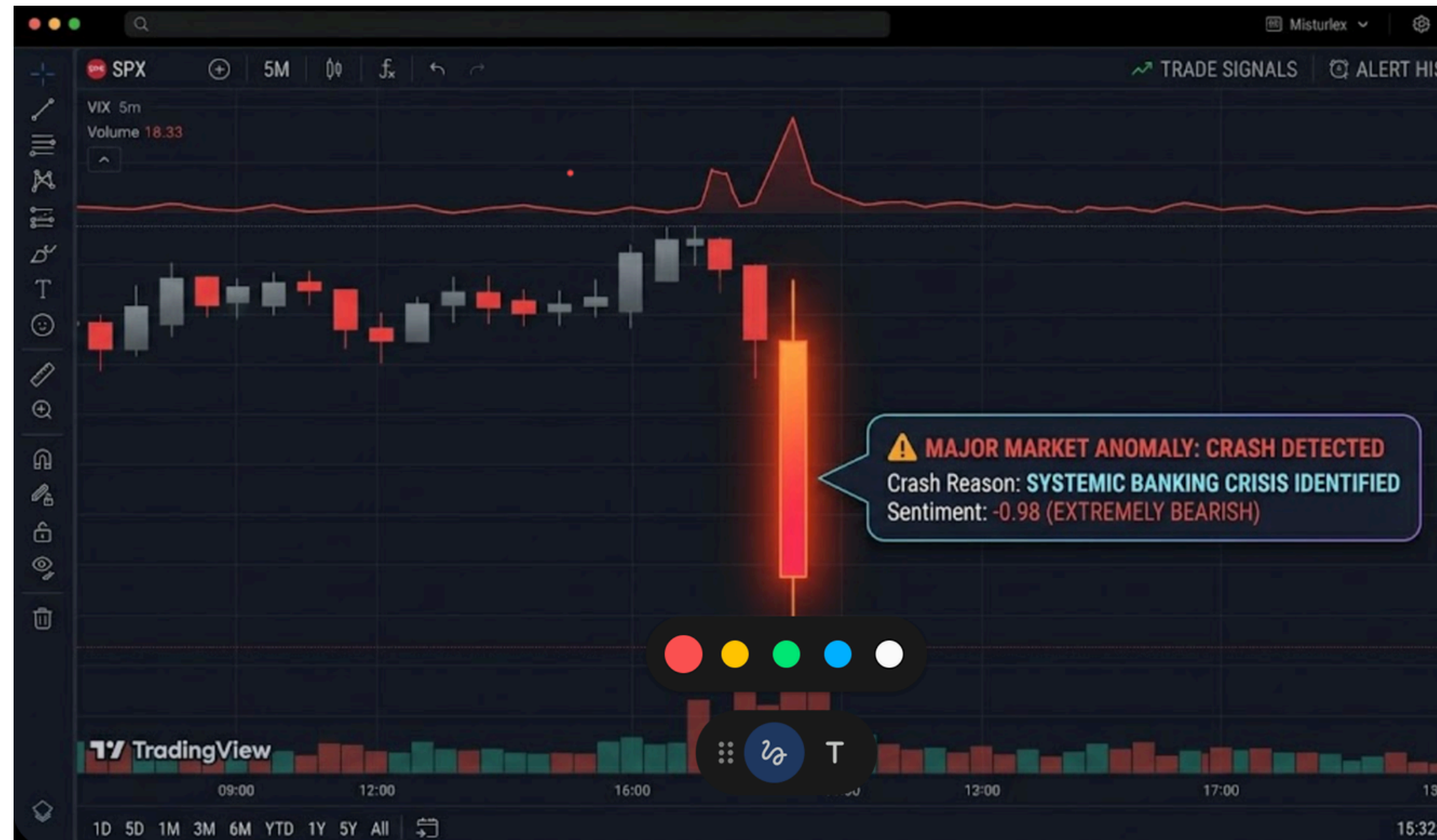
Milestone 3: Build the React frontend, integrate Lightweight Charts, and finalize the dark-mode UI/UX.

Phase 5 (Weeks 12-14)

System testing, latency optimization, and drafting the final academic research paper.

Conclusion

Latency and context issues in modern retail trading. By fusing statistical anomaly detection with state-of-the-art Natural Language Processing (FinBERT), this project automates the difficult task of market event attribution. The resulting system transforms raw, noisy market data into a clean, actionable, and visually intuitive interface, serving as both a powerful trading tool and a robust foundation for academic research in financial data science.



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- [9] R. Chen, H. Wang, and T. Zhao, "A hybrid LSTM-based approach for event-driven stock price anomaly detection," *Knowledge-Based Systems*, vol. 260, 2023.
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Thank you !